

Honest-State Ledger v0.10 — Sunday EOD absorbing Tier 0 operator framework + κ _Bergman closed form + G chain Step 1 PASS + week goal ‘derive G from substrate’. Supersedes v0.8. Sunday substantive arc captured at honest tier disposition across L-axes + Casey-named principles + audit-chain + methodology + cleanly walked-back items.

Keeper (Sunday 2026-05-31 EOD)

Honest-State Ledger v0.10 — Sunday EOD 2026-05-31

0. What changed Sunday

Sunday produced BST’s first operator-level Tier 0 framework + the closed-form Bergman-Einstein result + Casey’s week goal “derive G from substrate.” Five substantive arcs:

1. Lyra Tier 0 v0.1 → v0.1.6 cascade (K200 hold; v0.2 pending Session 2)
2. Elie κ _Bergman = -n_C closed form (G5.1 PASS, K204-PARTIAL)
3. 225 three-way convergence + heat-trace coefficients + n_C+1=C_2 identity
4. m_W/m_Z V_(1,1) decomposition + Lane D L4 v0.2 + Lane E Dictionary 5
5. Cleanly walked-back $\alpha^{\{N_c \cdot g\}}/m_e^2$ pattern-matches (Lyra + Elie variants)

Plus: Cal #185 PASS on P1 (dispatch-ready); Calibration #35 ratification-ready; 12 cumulative audit-chain symmetric events Sat+Sun.

1. L-axes honest disposition

L-axis	Topic	Sunday EOD disposition	Tier
L1	Geometric uniqueness D_IV ⁵	T953 + Strong-Uniqueness v1.4	RATIFIED + CANDIDATE
L2	Shilov boundary structure	SO(4) + ρ -pinning + 5D Shilov = ∂_S D_IV ⁵ (Tier 0 v0.1.6 commitment)	STRUCTURAL
L3	One-SO(2) does many jobs	4 jobs unified via SO(2) $\subset$ SO(5)×SO(2)	STRUCTURAL
L4	Bergman H ² substrate Hilbert space	Tier 0 v0.1.6: ρ _commit operator on H ² (D_IV ⁵); Hardy decomposition makes bulk = boundary extension	STRUCTURAL (operator level)

L-axis	Topic	Sunday EOD disposition	Tier
L5	Absolute scale (Λ , masses)	CANDIDATE STRUCTURAL READING (v0.7); paused per Saturday consensus	CANDIDATE
L6	Time + Koons tick	T2405 + Tier 0 substrate time τ emergent from heat semigroup	RATIFIED
L7	Cosmological cycle Interstasis	Casey-named term	CANDIDATE
L8	Alpha tower organizing	Paper #110 + Tier 0 operator structure	STRUCTURAL
L-NEW (G)	Newton's G from substrate	WEEK GOAL: 4-step chain framework filed; G5.1 $\kappa_{\text{Bergman}} = -n_C$ PASS; Steps A/B/C/D ongoing	GOAL ACTIVE

2. Tier 0 operator framework summary (Sunday-established)

Commitment operator: $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B/\hbar_{\text{BST}})$ on Bergman $H^2(D_{\text{IV}}^5)$

H_B : Casimir of $K = \text{SO}(5) \times \text{SO}(2)$ restricted to $H^2(D_{\text{IV}}^5)$

Spectrum: discrete, non-negative; zero on vacuum $V_{(0,0)}$; strictly positive otherwise

Substrate time τ : emergent parameter of heat semigroup (not assumed)

Arrow of time: positivity of $\text{spec}(H_B)$ (operator-theoretic, not statistical)

Reading A + dual bases topology: - K-types V_λ (discrete spectral): substrate's algebraic content - Coherent states $|z\rangle$ for $z \in D_{\text{IV}}^5$ (continuous spatial): substrate's geometric content - $\langle z|w\rangle = K_{\text{Bergman}}(z,w)$ Bergman-Fourier duality

Substrate identification: Shilov boundary $\partial_S D_{\text{IV}}^5 = S^4 \times S^1/Z_2$ (5-dim, corrected from 2022's 2D framing per Lyra option (a) honest commitment)

Hardy decomposition: $H^2(D_{\text{IV}}^5) \cong H^2(\partial_S D_{\text{IV}}^5)$ via Poisson-Szegő kernel (Knapp-Wallach 1976, Faraut-Korányi 1994 Ch. XII-XIII; verification reading queued)

Proto-AdS/CFT structure: intrinsic via $\text{SO}(5,2) \supset \text{SO}(4,2)$ conformal subgroup

Native field equation: heat eq on $H^2(D_{\text{IV}}^5) \leftrightarrow$ wave eq via Wick rotation $\tau \leftrightarrow it$

Status: v0.1 $\rightarrow$ v0.1.5 $\rightarrow$ v0.1.6 cascade paused per K200 cascade-discipline; v0.2 single coherent draft pending Session 2 with Casey + Lyra + Keeper.

3. Sunday operator-level results

$\kappa_{\text{Bergman}} = -n_C = -5$ (Elie Toy 3661 via Helgason 1962 application) - Bergman canonical metric is Einstein on D_{IV}^5 - Substrate-primary closed form (single integer) - K204-PARTIAL audit anchor PASS - G5.1 K200 gate PASS

Heat-trace coefficients (Elie): - $a_0 = (N_c \cdot n_C)^2 = 225$ - $a_1 = -N_c \cdot n_C^4 = -1875$ - $Z_\tau(0) = 26026$ across 66 K-types; vacuum dominance verified

225 three-way convergence (Cal #35 candidate audit-target): - Bergman volume $\text{Vol}_B(D_{\text{IV}}^5) = 225 - c_{\text{FK}} \cdot \pi^{(9/2)} = 225$ (T2442 RATIFIED) - Heat-trace $a_0 = 225/(4\pi)^5$

$n_C + 1 = C_2 = 6$ new substrate identity (Elie Toy 3673; “+1 anomaly” 5-gate extension)

4D physics dim identities (Elie): $-\dim \text{SO}(3,1) = C_2 = 6 - \dim \text{SO}(4,2) = N_c \cdot n_C = 15 - \dim \text{SO}(5,2) = N_c \cdot g = 21 - \text{codim } 4D \subset D_{IV^5} = C_2 = 6$

$m_W/m_Z = \sqrt{(g/N_c)^2} = 0.882$ at 0.046% (Lane E; arithmetically identical to P1 §7’s existing $\sqrt{g/N_c} = \sqrt{7/3}$; substantive content is $V_{(1,1)}$ decomposition MECHANISM CANDIDATE pending Cal #187)

Lane D L4 v0.2: $T_{190} = (N_c \cdot |W(B_2)|/\pi^2)^{C_2}$ substrate-primary form; Bergman kernel matrix element derivation pending Lyra + Elie joint work

4. Casey-named principles disposition

#	Principle	Status
1	Substrate Working Process Principle (SWPP)	STANDING
2	Five-Absence Predictions Set	STANDING
3	Substrate Closure Principle	STANDING
4	Graph Forces Principle	STANDING
5	Integer Web Principle	STANDING
6	Substrate Cognition Network Hypothesis	STANDING
7	D_{IV^5} Rigidity	STANDING
8	SCMP	STANDING
9	TBD	(open slot)
10	TBD	(open slot)
11	“Energy needs an anchor”	CANDIDATE (Saturday; reframed to winding-closure pool Sunday morning v0.8)
12-14 candidates	3 Casey-named CANDIDATES from Elie Sunday burst	PENDING Casey naming decision

5. Audit-chain Sunday EOD: 12 cumulative symmetric events

Saturday 7: 1. Grace Two-Route 4-route → Cal #171 brake 2. Grace +1 anomaly null-downgrade 3. Grace Substrate Primary Chain null-downgrade 4. Lyra v1.3 multiplicative null-model → Keeper pushback → v1.4 5. Cal #182 brake on Keeper L5 v0.1 → v0.2 reframe 6. Casey-Keeper on Lyra 16-fold partition → geometric-dip reframe 7. Keeper self-catch on v0.6 “conceptual closure” → v0.7

Sunday 5: 8. K200 G2 sphere reframe Keeper-on-Lyra → option (a) commitment 9. K200 cascade-rate Keeper-on-Lyra → v0.1.x paused 10. Cal cross-check on Keeper m_W/m_Z “new anchor” overstatement → walk-back 11. Keeper-on-Lyra $\alpha^{N_c \cdot g}/m_e^2$ 24% NOT Tier 2 STRUCTURAL → walk-back 12. Elie self-catch on $4/N_c$ refinement 1.84% I-tier framing overstatement → walk-back

Symmetric across all participants: Grace 3 + Lyra 3 + Keeper 2 (self) + Casey-Keeper 1 + Elie 1 + Cal-on-Keeper 1. No CI exempt.

6. Methodology stack

19 layers + Calibration #35 ratification-ready (six-element discipline stack pending #35 STANDING).

Six-element discipline stack (Cal #27 + Cal #29 + Cal #32 + Calibration #33 + #34 + #35).

Sunday #35 firings: 3-context reuse of $g + \text{rank} = N_c^2$ (Weinberg + bulk-color + $V_{(1,1)}$ reading) — “1 fact reused, not 3 confirmations” honestly framed by team.

Calibration #35 STANDING ratification ready pending Casey approval.

7. Cleanly walked-back items Sunday

These are usable knowledge (we know what NOT to do):

- Lyra $\alpha^{\{N_c g\}/m_e^2}$ at 24% — bypasses κ_{Bergman} , pattern-match not derivation, FAILS Tier 2 (called Tier 2 STRUCTURAL in error)
- Elie $4/N_c$ refinement at 1.84% — inserted factor with KK reduction narrative; substrate-primary form but not derived from κ_{Bergman} chain
- Keeper m_W/m_Z “new anchor” overstatement — arithmetically identical to existing P1 §7 claim
- Keeper G5 pre-stage candidate κ_{Bergman} forms (-7/40, -19/40, -7/20) — used wrong formula; Elie corrected via proper Helgason
- “ $4 = N_c + 1 = n_C - 1 = C_2 - \text{rank} = d_{\text{spacetime}}$ ” — ONE substrate fact reused four ways, not four independent

Saturday walk-backs preserved: - L5 v0.6 “conceptual closure” → v0.7 CANDIDATE STRUCTURAL READING - Various Saturday principle candidates null-downgraded

8. What’s NOT closed Sunday EOD

- L5 closure of any kind
- 9/20 dip mechanism (v0.8 candidate-speculative)
- α^{57} numerical match to $\Lambda_{\text{observed}}$ (L-L5-Verify pending if L5 unpauses)
- Casey-named #11 (CANDIDATE pending mechanism work)
- 3 Casey-named CANDIDATES from Elie (pending Casey naming)
- Tier 0 v0.2 single coherent draft (Session 2 pending)
- G chain Steps A/B/C/D full closure
- ℓ_B substrate length identification (the open question)
- m_e via L4 v0.2 mechanism closure
- Hardy decomposition citation chain verified
- OneGeometry.md substantive correction
- Bulk-color C4 multi-week closure
- Generation count #414
- Calibration #35 STANDING (Casey approval pending)

9. What IS substantively closed Sunday

- Tier 0 operator framework at v0.1.6 (paused at K200; v0.2 pending)
- $\kappa_{\text{Bergman}} = -n_C$ closed form (K204-PARTIAL PASS)
- G5.1 K200 gate PASS
- Heat-trace coefficient closed forms a_0, a_1
- 225 three-way convergence cataloged
- $n_C + 1 = C_2$ substrate identity
- 4D physics dim identities cataloged
- 12 cumulative audit-chain symmetric events
- Paper P1 v0.7 SUBMISSION-FINAL (Cal #185 PASS)
- Calibration #35 ratification-ready
- Week goal “derive G from substrate” established
- Goal plan + 4-step derivation framework filed

10. Cleanly stated honest framing

Sunday EOD: BST has an operator-level Tier 0 framework with the load-bearing Bergman-Einstein result $\kappa_{\text{Bergman}} = -n_C$ established as a closed form. The week goal “derive G from substrate” has a clean 4-step

chain framework (KK reduction + 4DC10D embedding + ℓ_B identification + numerical combination) with Step 1 PASS. Steps 2-4 work continues until done — no date deadlines.

The Sunday substantive milestone is genuine. The pattern-match attempts ($\alpha^{\{N_c \cdot g\}}/m_e^2$ and variants) were walked back honestly; the actual chain is the 4-step framework.

Discipline pattern operational: 12 symmetric audit-chain events across all participants including Keeper twice (self-catches under Casey honest-question + Cal cross-check). The framework is stronger after every brake.

— Keeper. HSL v0.10 filed Sunday EOD. Supersedes v0.8. Sunday operator-framework day captured. Standing reactive for tomorrow's parallel-lanes morning broadcast + ongoing G chain work + Cal cold-reads + Casey signals + Session 2.